

Muhammad Hammad Maqsood

AI / LLM Engineer · Agentic Systems · RAG · LLM Evaluation

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SUMMARY

AI/LLM engineer with end-to-end feature ownership across four production AI products at IgniteTech (GFI Software). Builds agent pipelines on LangGraph, hybrid retrieval-augmented generation over enterprise documentation, and the evaluation harnesses that decide whether any of it ships. Python is the primary language across all of it, and the working habit is to measure first, recommend second, and file the residual risk rather than hide it.

CORE SKILLS

Agents & Orchestration: LangGraph | LangChain | multi-agent routers, shared state & streaming | ReAct and planner-executor (ReWOO) agents | Model Context Protocol (MCP) | OpenRPC progressive disclosure | agentic workflow design

RAG & Retrieval: RAG | hybrid BM25 + vector search | Reciprocal Rank Fusion | vector database design (pgvector, FAISS) | PostgreSQL tsvector | chunking & overlap tuning | embeddings (sentence-transformers, Amazon Titan, OpenAI) | page-cited grounded answers

Evaluation & Observability: LLM-as-judge harnesses | blinded A/B with position randomisation | execution-grounded evals | hallucination testing and guardrails | production monitoring and alerting | Langfuse tracing and datasets | CloudWatch Logs Insights | COGS-per-query analysis | MLflow | SHAP

LLM Integration & Tooling: Anthropic Claude | OpenAI GPT | Google Gemini | AWS Bedrock | tool / function calling | schema-driven tool generation | structured outputs | prompt engineering | context engineering | fine-tuning (QLoRA/Unsloth, SFT, DPO) | Ollama

Cloud & Data: Python | TypeScript | React | SQL | AWS Lambda, Step Functions, SQS/SNS, EventBridge, S3, ECS/Fargate, ALB | PostgreSQL | MongoDB | FastAPI | Django/DRF | Docker & docker-compose | GitHub Actions CI/CD | PySpark | multi-tenant isolation | OIDC

EXPERIENCE

AI/ML Engineer · IgniteTech (GFI Software)

Nov 2024 – Present

Remote. End-to-end feature ownership across four production AI products: AppManager AI, AppManager RADAR, Gensym G2 Copilot, iMCP

- Designed and shipped AppManager AI's multi-node LangGraph conversation pipeline (help/main/summarize/query-rewrite agents, routers, shared state, streaming) across **47 files**, still the documented product architecture, plus a schema-driven ToolFactory that auto-generates LangChain tools from OpenAPI-style schemas with retrieval-based tool selection, behind the product's FastAPI backend (SSE streaming chat, auth-init gate, admin hot-reload endpoint). Built the provider-agnostic LLM layer under it, one streaming interface across AWS Bedrock, OpenAI and Google so model swaps are configuration-level, shipped with a verified failover trace.
- Built the Gensym G2 documentation copilot's retrieval service: hybrid BM25 over PostgreSQL tsvector fused with pgvector by Reciprocal Rank Fusion, returning page-cited answers across **3 manuals, 3,408 pages indexed as 3,511 searchable chunks**, shipped as a customer-facing MCP server with per-customer HMAC-SHA256 keys and a split-execution local proxy, TypeScript on Node, published to npm as @ignitotech/gensym-mcp, and migrated the packaged inference stack off LlamaIndex onto LangChain to fix a FAISS format incompatibility.
- Built the evaluation infrastructure the product decisions rest on: an execution-grounded G2 codegen harness (**20 tasks, 6 categories, 135+ tests**) run against a live instance to establish the pre-fine-tuning baseline for the frontier model under test at **95% (19/20)**, and a blinded LLM-as-judge bake-off that returned an honest NO-GO on a model measured 74.7% cheaper because the incumbent won 60 head-to-head against 18.
- Kept that evaluation honest against itself rather than defending it: flagged that the offline correctness heuristic was penalising the challenger purely for answering roughly twice as long when the extra content was accurate, and treated the judge model (stronger than either contestant) as one vote beside an appliance-link audit and independent trace verification.
- Built AppManager AI's application-layer observability and error monitoring: structured logging, a two-tier ERROR/CRITICAL severity convention and full tracebacks on every hot-path handler, and specified the CloudWatch to Lambda to SNS to Slack routing that gave a live customer beta its first real-time alerting. Set the company's official COGS-per-query baseline over **174 production traces**, revising a prior estimate.

- Hardened tool calling under live customer load: coerced malformed tool arguments at the LangGraph ToolNode boundary to end token-burn recursions that made a single failed conversation cost a multiple of a normal one, cut a disconnected-appliance hallucination loop from **7 retries to exactly one call**, and collapsed 10 per-product article tools into one enum-constrained dispatcher (10/10 unit and 6/6 live scenarios), and threaded MSP account scoping through **9 tools** in a three-layer isolation design, **25 regression tests**, verified on a production MSP org.
- Root-caused silent SSE stream disconnects to an AWS ALB 60s idle timeout and shipped a heartbeat fix, plus a 24h eligibility cache with ALB stickiness and a malformed-turn recovery with localised fallback across the React chat widget in **8 languages**.
- Wrote the RCA, impact analysis and corrected quantification that answered Central Engineering's objection that push ingestion would transfer risk rather than resolve it, unblocking architectural approval; then designed, implemented and end-to-end validated the replacement, a single OIDC-protected push call in place of **7+ fetch calls per appliance**, with the companion alert-type integration merged to production by that team.
- Authored the core iMCP standard, a tiered certification framework with separate MCP client and server integration checklists, validated against two reference implementations. Built the Kerio Control A(I)PI layer on it, exposing a **334-method** JSON-RPC admin API through only **3 tools** (OpenRPC progressive disclosure) so the tool surface never overloads the context window, on a stale community framework forked, updated and republished to npm as openrpc-mcp-server-updated, then attached it to a LangGraph ReAct agent that turned generic recommendations into data-grounded investigations across a **28-anomaly** batch at ~2 min each, **23 of 28** persisted.

CureMD

Oct 2022 – Nov 2024

AI Engineer (Oct 2023 – Nov 2024) · Associate AI Engineer (Apr 2023 – Sep 2023) · Trainee AI Engineer (Oct 2022 – Apr 2023). Regulated healthcare ML

- Built the Medlabs chronic-disease prediction pipeline: PySpark feature engineering over **593,055 patients across 547 US primary care centers** into a **~40-feature** modelling set, persisted to MongoDB as the feature store for both training and the downstream clinical apps, then a multi-label gradient-boosting classifier paired with a rule-based confirmation layer and served through a Flask validation app. Published as a first-author paper, arXiv:2603.14876.
- Delivered the SMART EOB system end to end to automate Explanation-of-Benefits posting at **~3,500 claims/day**, with **~6x faster** batch OCR, roughly a third of the previous human posting effort, and ~70% of claims auto-detected as already posted so reviewers only handled the rest.
- Engineered the RabbitMQ + Celery parallel pipeline behind it (chains and chords, worker counts tuned to GPU and DB concurrency limits), a dual-engine OCR extractor aligning PaddleOCR and Tesseract on the same text line by coordinates, and a reusable FastAPI OCR microservice deployed on Linux via Nginx and Unicorn.

Trading Automation Developer · MetaQuotes Ltd, then freelance (MQL5, Upwork)

Apr 2020 – Apr 2023

MetaQuotes engagement Apr 2020 – Jan 2022. Independent client work on Upwork continued to Apr 2023

- Designed trading strategies with clients and automated them in MQL4/MQL5 for the MetaTrader platforms. Platform-verified delivery record: **41 of 41 MQL5 jobs completed**, rated **4.7 out of 5 across 26 client ratings**, and **12 of 12 jobs completed on Upwork**.
- Moved from MQL into Python on paid client work: built a KuCoin crypto trading bot consuming WebSocket market-data streams to drive automated trade decisions (Upwork, Feb – Apr 2023), and a Django feature making a collection of forms editable on a single page.

PUBLICATIONS

A Hybrid AI and Rule-Based Decision Support System for Disease Diagnosis and Management Using Labs.

arXiv:2603.14876, March 2026. First-author. Earlier version presented at the AIME Real-World-Evidence workshop, July 2024.

CERTIFICATIONS

Most relevant of 18 completed certifications

Functions, Tools and Agents with LangChain (DeepLearning.AI) · Building and Evaluating Advanced RAG (DeepLearning.AI) · LangChain: Chat with Your Data (DeepLearning.AI) · Generative AI with Large Language Models (DeepLearning.AI) · Create AI Powered Apps with Open Source LangChain (IBM Skills Network) · ChatGPT Prompt Engineering for Developers (Coursera) · Machine Learning Specialization (Stanford University)

EDUCATION

Bachelor of Engineering (BEng), Mechanical Engineering — National University of Sciences & Technology (NUST), Pakistan — 2017–2021